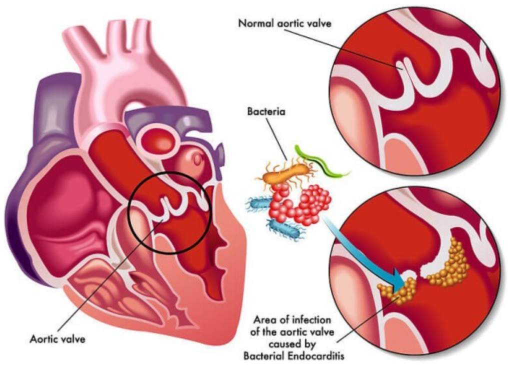
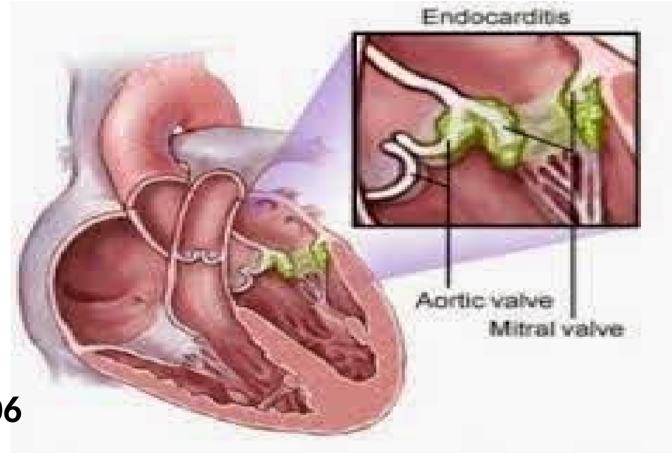


Infective Endocarditis

Dr. Hanaa Elsayed



Kumar & Clark's Clinical Medicine 10th edition page 457 edition page 1103 -1106

Learning outcomes

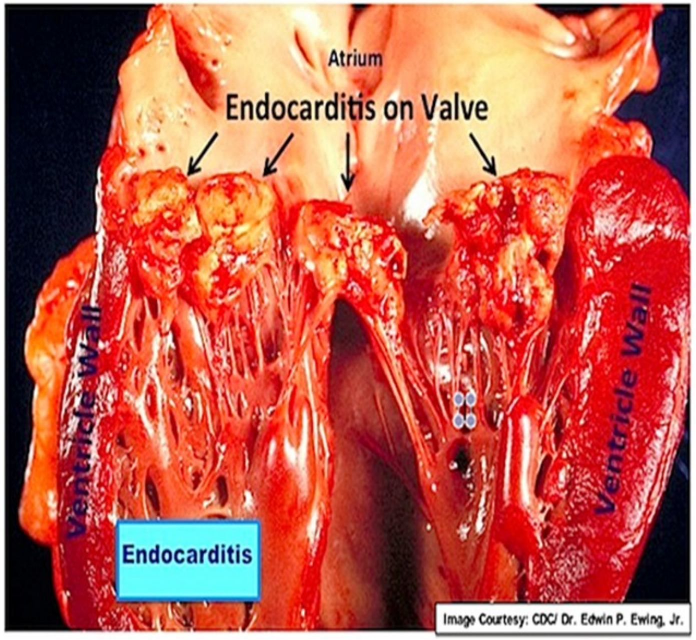
- **Define infective endocarditis (IE)**
- **Enumerate Risk factors of IE.**
- **List Clinical pictures of IE.**
- **Describe how to Diagnose IE.**
- **Construct Treatment of IE**

Infective endocarditis (IE)

- **It is a serious infection affect the innermost layer of the heart, atrial and ventricular endocardium, that often involves valves (normal and/or prosthetic).**
- **It affect one or more valves (Aortic and mitral most common) and associated with destruction of cardiac tissue.**
- **Right side valve in IV drug users.**
- **IE affects 1-5/ 100 000 population/ year.**
- **IE is 3 times more common in males especially elderly.**

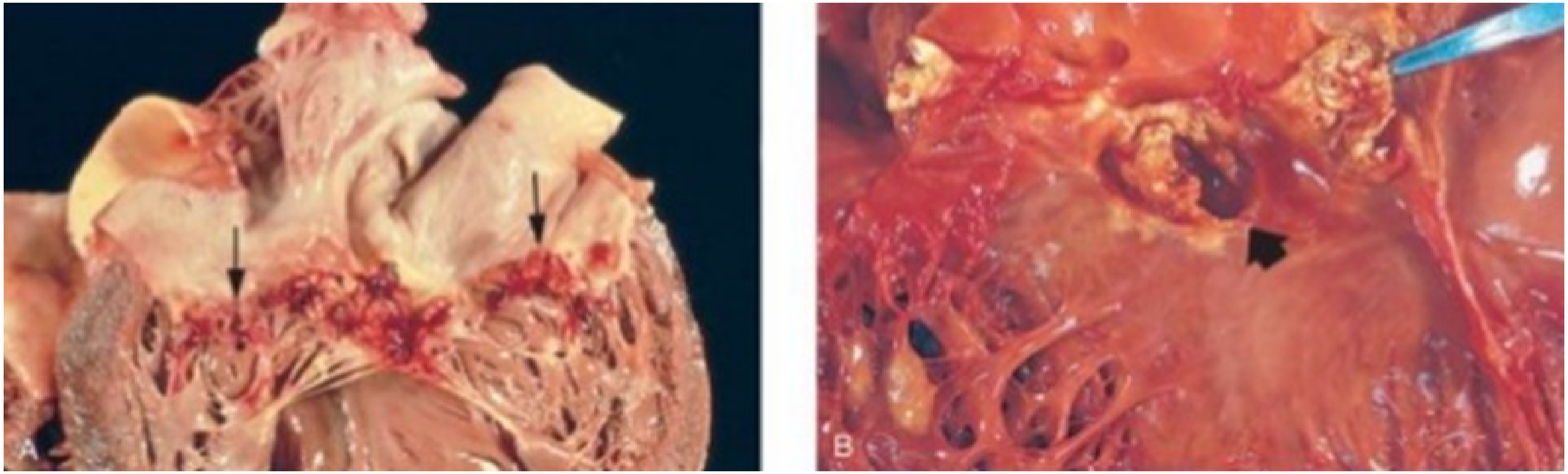
Risk factors

- **Cardiac causes:**
  - **Prosthetic heart valve.**
  - **Previous history of IE.**
  - **Congenital and rheumatic heart disease.**
  - **Degenerative valve disease.**
  - **Cardiac transplant with valvulopathy.**
  - **Implantable electronic cardiac device (pacemaker).**



- Non-cardiac
  - Hemodialysis.
  - Diabetes mellitus.
  - Intravenous drug use.
  - Immunosuppression.
  - Poor oral hygiene.
  - Dental treatment.
  - No identifiable cause in 25-47%

- # Etiology
  - Endocarditis is usually the consequence of two factors:
    - The presence of organisms in the bloodstream.
    - Abnormal cardiac endothelium that facilitates their adherence and growth.



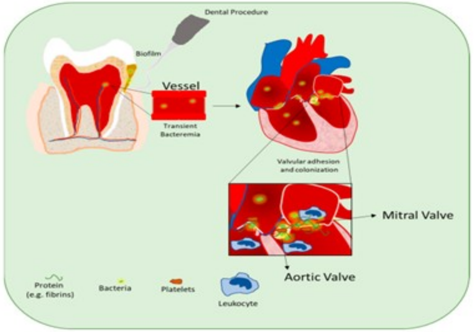
Causative organisms

- **Bacterial (98%):**
  - **Viridans streptococci (subacute) (35-60%).**
  - **Staphylococcus aureus (acute) (30-40%) IV drug abuse, long standing IV catheters, Diabetics.**
  - **Enterococci (8%) more in elderly males.**
  - **Gram negative bacilli 5%: klebsiella, pseudomonas and brucella.**

- **HACEK organism:**
  - **Hemophilus, actinobacillus, cardiobacterium, eikenella, kingella.**
- **Fungal (2%): Candida albicans in IV drug users and immunocompromised.**

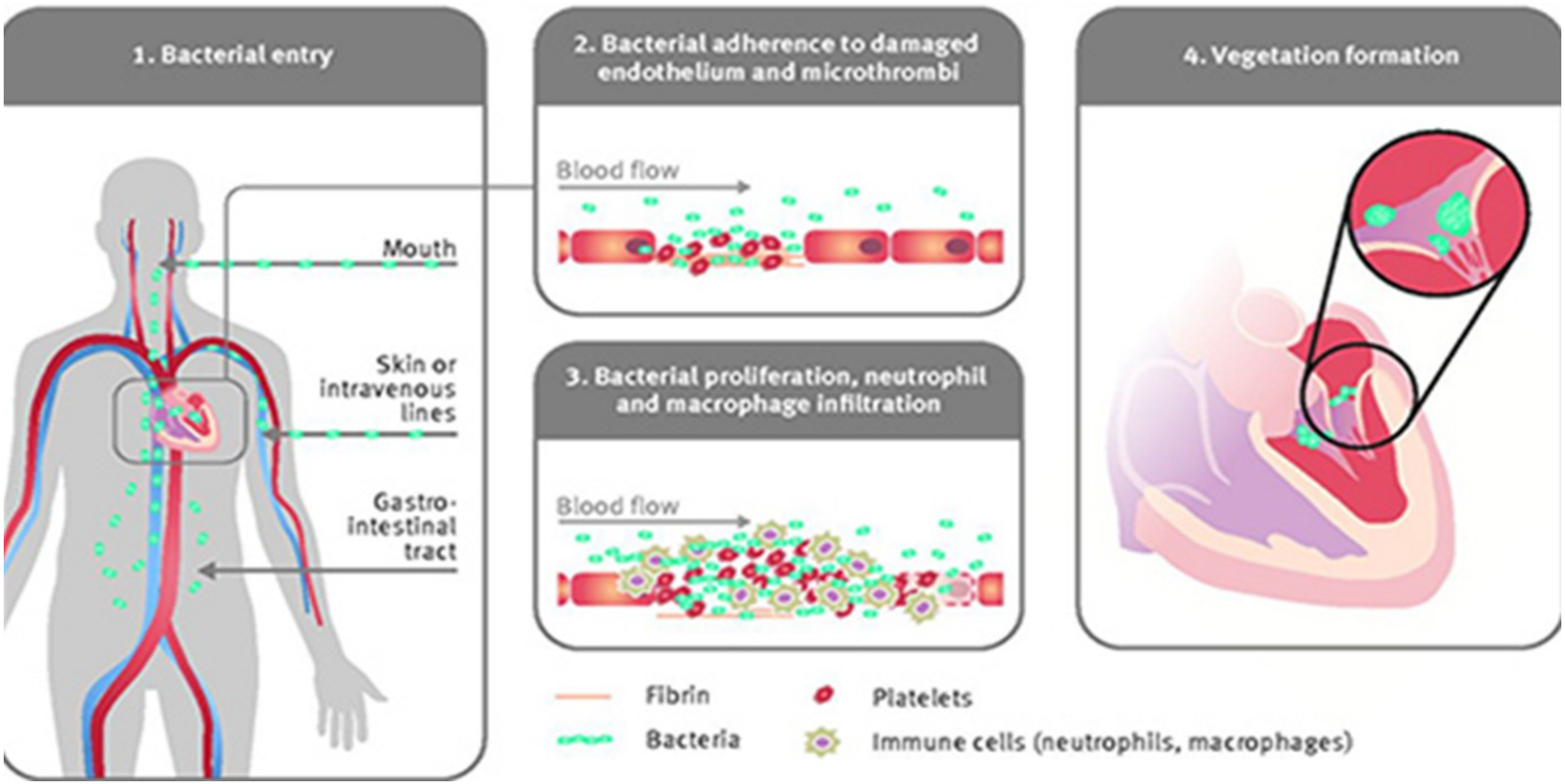
Route of infection

- **IE occurs when bacteria enter bloodstream through:**
  - **Mouth:**



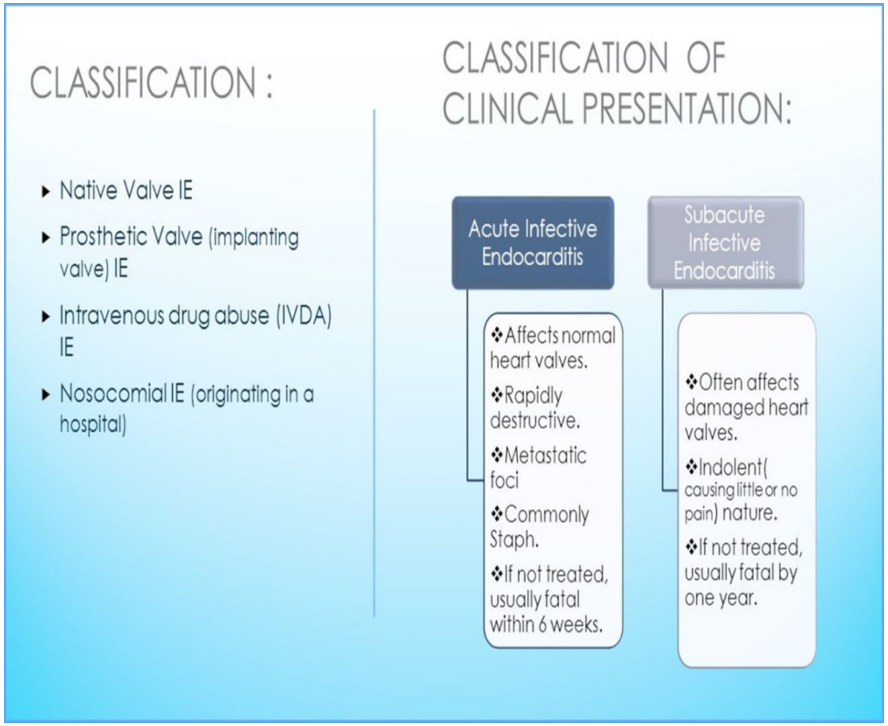
    - **Dental disease or procedures.**
    - **Poor dental hygiene**
    - **Causative organism: Streptococcus viridans.**
  - **Gastrointestinal (Gut) and genitourinary disease or procedures: Usually strept. Foecalis or gram –ve bacilli.**
  - **Skin and soft tissue infections:**
    - **Intravenous (IV) cannula, IV drug use, cardiac surgery or catheterization).**

- Damaged endocardium promotes platelet and fibrin deposition, which allows organisms to adhere and grow, leading to an infected vegetation (thrombotic debris and organisms).



Classification

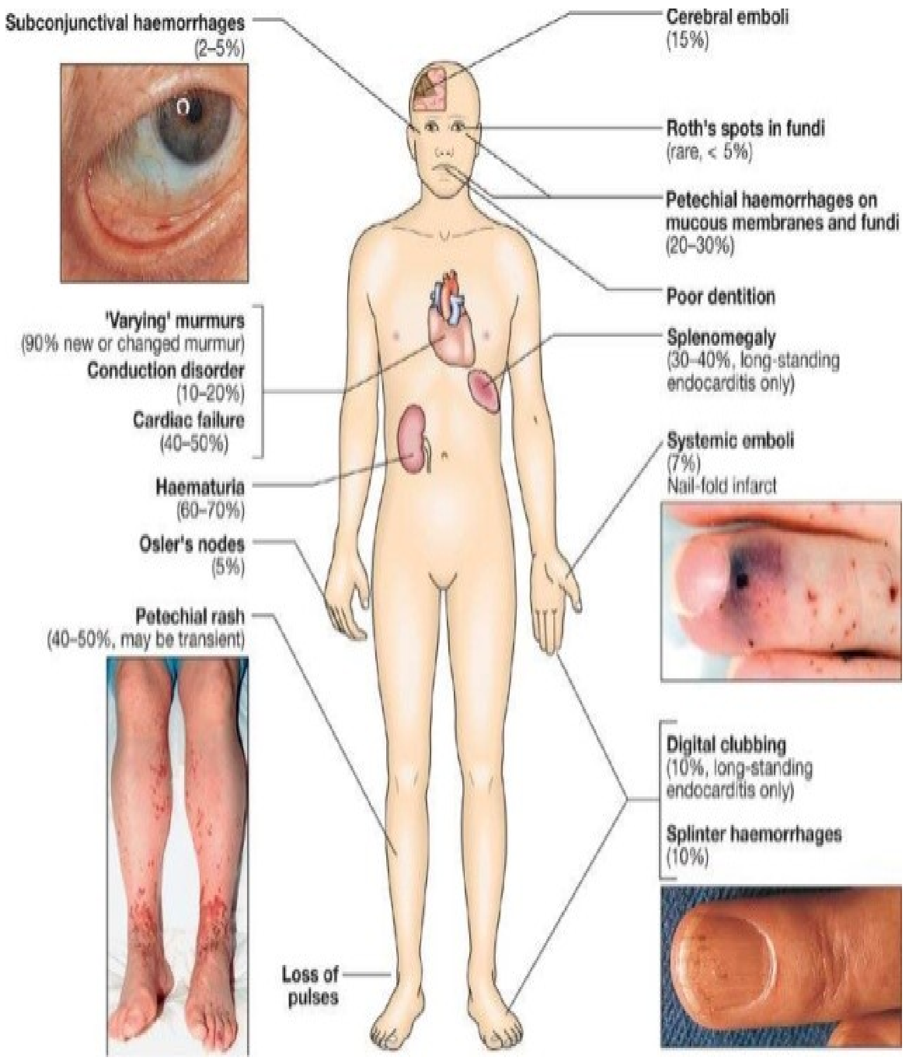
- **I-Non-infective endocarditis:** **e.g.** **rheumatic fever, systemic lupus erythematosus (SLE).**
- **II- Infective endocarditis:**
- **According to causative organism:**
  - **Bacterial endocarditis.**
  - **Non bacterial endocarditis (viral, rickettsial or fungal).**



- **According to clinical presentation:**
  - **Acute bacterial endocarditis (few days to weeks):**
    - **Acute onset, rapidly progressive course, affecting disease or even normal hearts, caused by highly virulence organism (Staphylococcus aureus and group B streptococci).**
    - **If untreated leads to death within weeks.**

- **Sub-acute bacterial endocarditis (few weeks to months):**
  - **Insidious onset, affecting diseased hearts, caused by low virulence organism (Alpha-hemolytic streptococci or enterococci).**
  - **Recover after appropriate antibiotic treatment.**
- **Chronic IE: with low grade fever and non-specific symptoms**

Clinical pictures



Malaise (95%), anorexia (loss of appetite), loss of weight, fatigue and headache.

Marked pallor, toxic face

- Fever >95%
- Arthralgias and/or myalgias 25-45%
- Murmur >85%
- Splenomegaly 25-60%
- Petechiae 20-40%
- Splinter hemorrhages 10-30%
- Roth's spots <5%
- Osler's nodes 10-25%
- Janeway lesions <5%
- Clubbing 10-20%
- Clinically apparent emboli 25-45%
- Neurologic manifestations 20-40%

Osler's Nodes

1. More specific
2. Painful and erythematous nodules
3. Located on pulp of fingers and toes
4. More common in subacute IE

Janeway Lesions

1. More specific
2. Erythematous, blanching macules
3. Nonpainful
4. Located on palms and soles

Splinter Hemorrhages

1. Nonspecific
2. Nonblanching
3. Linear reddish-brown lesions found under the nail bed
4. Usually do NOT extend the entire length of the nail

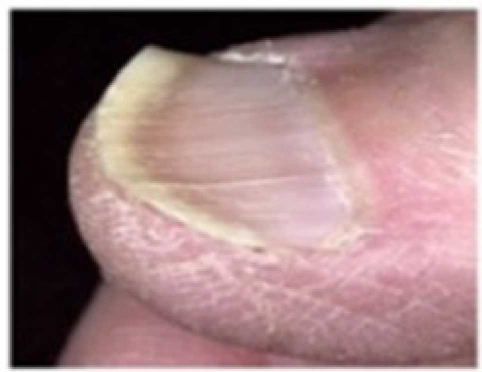
Pulse: tachycardia, or absent due to embolization.

Pale clubbing (from 2nd to 3rd week).

Nail Clubbing

Hand signs

Clubbing

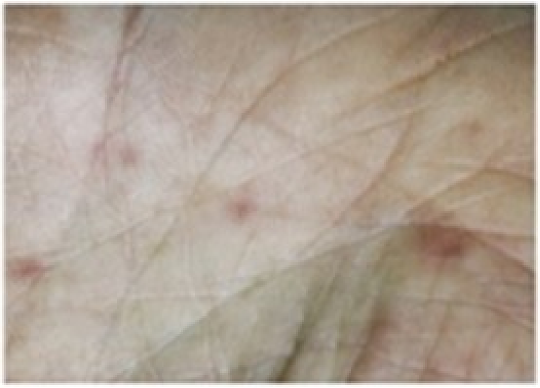


Koilonychia

Palmar erythema

Splinter haemorrhages

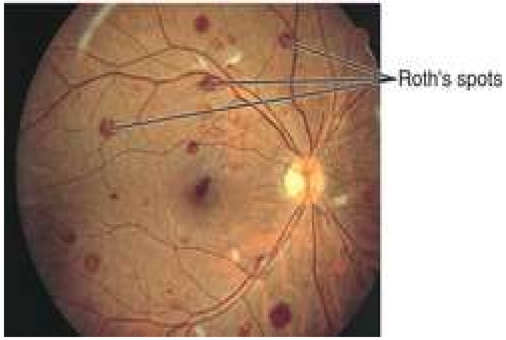
Osler nodes

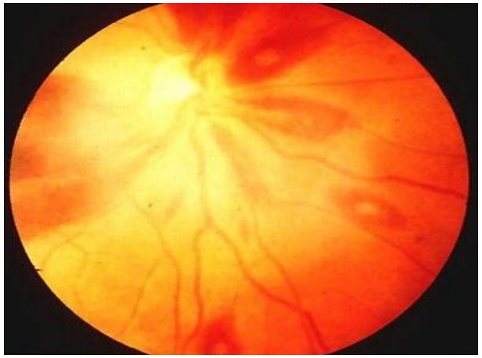


Janeway lesions

– Roth spots in the retina (retinal hemorrhages with white or pale centers).

– Sudden blindness due to central artery embolism.

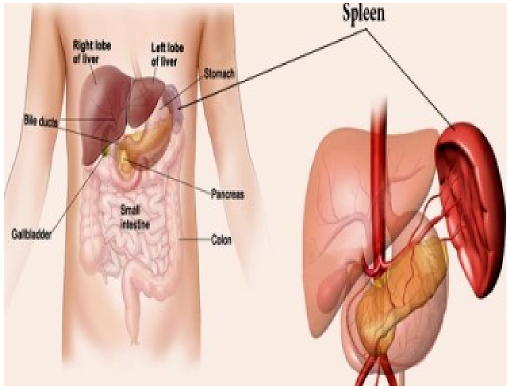




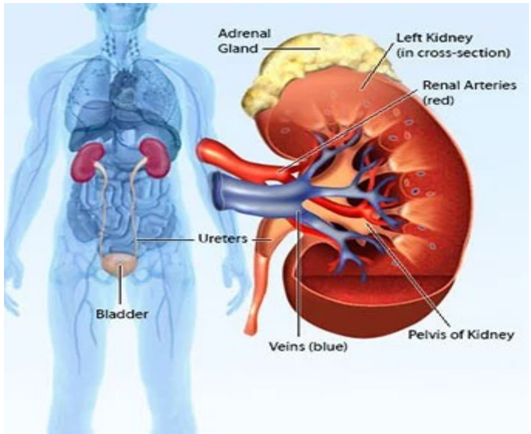
Subconjuctival Hemorrhages

- **Cardiac manifestation:**
  – Features of underlying cardiac disease and heart failure as dyspnea.
  – Heart murmurs (new or changing).

- **Other organs:**
  - **Spleen:**



    - **Slight enlarged, soft and tender.**
    - **Stitching pain due to splenic infarction, perisplenitis.**
  - **Kidney:**
    - **Glomerulonephritis (GN), hematuria.**



  - **CNS:**
    - **Hemiplegia (cerebral emboli).**

Prognosis and complications

- **Mortality rate 20-25% and morbidity (50-60%) despite the use of antibiotics.**
- **Complications:**
  - **Heart failure, myocardial abscesses, myocarditis and arrhythmias.**
  - **Embolization.**
  - **Renal: GN, renal failure.**
  - **Neurologic manifestations:**
    - » **Brain abscess, hemiplegia, subarachnoid hemorrhage, due to rupture of mycotic aneurysm of circle of Willis.**
  - **Complication of treatment.**

Modified Dukes Criteria for diagnosis of IE

- **A definitive diagnosis:**
  - 2 major criteria Or
  - 1 major and 3 minor Or
  - 5 minor criteria.
- **Possible:**
  - 1 major and 1 minor Or
  - 3 minor.

Major Criteria	
Positive blood cultures Typical pathogens from at least two separate cultures	
Evidence of endocardial involvement by echocardiography Endocardial vegetation, perivalvular abscess, new partial dehiscence of prosthetic valve, new vavular regurgitation	
Minor Criteria	
Predisposition Heart condition or IV drug use	**Fever** Greater than or equal to 38°C
Microbiologic evidence Single positive blood culture (except for coagulase-negative staphylococcus or an organism that does not cause endocarditis)	**Vascular phenomena** Arterial emboli, mycotic aneurysm, septic pulmonary infarcts, conjunctival hemorrhages, Janeway lesions
Echocardiographic findings Consistent with endocarditis, but does not meet major criteria	

Major criteria

- **Positive blood culture:**
  - **2 blood cultures positive for organisms. or**
  - **3 or more positive cultures drawn (first and last) at least 1 hour apart. or**
  - **Persistently positive cultures more than 12 hours apart. or**
  - **Serological test positive for Q fever, immunoglobulin G antibodies (immunofluorescence assay) at a titre >1 : 800 or**

- **Major echocardiographic criteria include:**
  - **An oscillating intracardiac mass on valve or on implanted material.**
  - **Myocardial abscess.**
  - **New partial dehiscence of a prosthetic valve**
  - **New murmur (regurgitation)**

Minor criteria

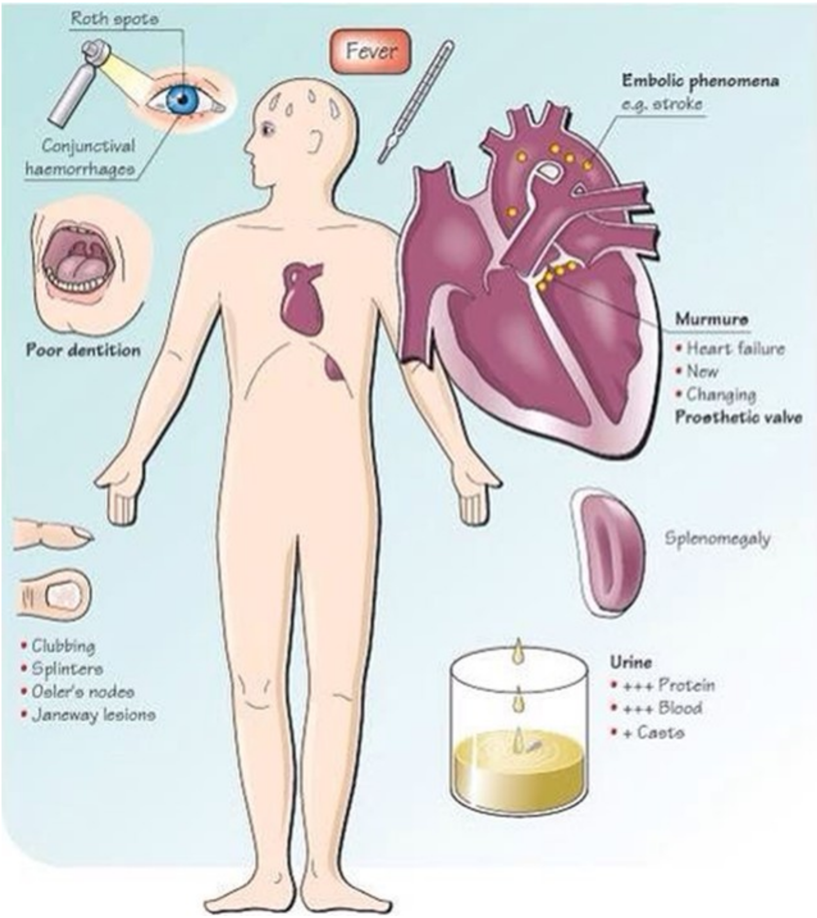
- **Predisposing heart condition or intravenous drug use.**
- **Fever: temp. >38°C (100.4°F)**
- **Vascular phenomenon, including:**
  - **Major arterial emboli.**
  - **Septic pulmonary infarcts.**
  - **Intracranial hemorrhage.**
  - **Conjunctival hemorrhage.**
  - **Janeway lesions.**

- **Immunologic phenomenon:**
    - **GN.**
    - **Osler nodes.**
    - **Roth spots.**
- **Positive blood culture not meeting major criteria.**
- **Echocardiogram:**
    - **Consistent with IE but not meeting major echocardiographic criteria.**

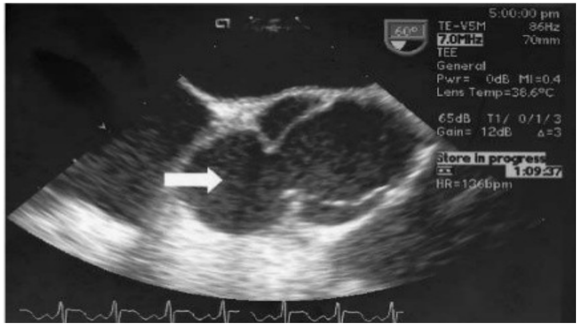
Investigation

- **Blood culture: confirm diagnosis of IE.**
  - **At least 3 sample from different venipuncture sites over 24 hours.**
- **Serology if blood culture negative (Culture negative endocarditis) (5-10%) include:**
  - **Q fever caused by coxiella burnetii**
  - **Chlamydia species, Bartonella and legionella.**
- **CBC:**
  - **Anemia (decreased hemoglobin level), increased WBCs, increased or decreased platelets.**

- CRP, ESR increased.
- Kidney function test (Urea and creatinine):
  - May elevated due to glomerulonephritis (GN).
- Liver biochemistry:
  - Serum alkaline phosphatase may be increased.
- Urine:
  - Proteinuria and microscopic hematuria.



- **ECG to detect any abnormality:**
  - **Myocardial infarction (MI).**
  - **Conduction abnormalities as prolonged PR interval.**
- **Echocardiography (Transthoracic and transoesphageal).**
  - **It is a non-invasive image.**
  - **More sensitive to detect presence and size of vegetation, abscess.**
  - **Assess cardiac function.**
  - **A negative echocardiogram does not exclude a diagnosis of endocarditis and CT-PET may be required, particularly in suspected prosthetic valve infection.**
  - **Chest x-ray.**



Treatment

- **Blood cultures should be taken prior to empirical antibiotic.**
- **Antimicrobial therapy: as soon as possible.**
  - **Use adequate dosage, Parenteral route.**
  - **Sufficient duration: 4-6 weeks or longer**
  - **Avoid change of antibiotic dosage or regimen unless there are culture.**
  - **Resolution of fever in 48 hours of initiation of antibiotic.**

- **If fever persist, patient should be evaluated for:**
  - **Extra cardiac abscess**
  - **Hospital acquired infection.**
  - **Drug reaction**
  - **Pulmonary embolism (secondary right-sided endocarditis or prolonged hospitalization).**
- **Combination therapy most important for:**
  - **Shorter course regimens.**
  - **Eradicating microbial infection and minimizing resistance.**

Antibotic regimen for infective endocarditis

- Streptococci
  - Benzyl penicillin (1.2g 4 hourly) 4-6 weeks
  - Gentamicin (1mg/kg 8-12 hourly) 4-6 weeks
- Enterococci
  - Ampicillin sensitive
    - Ampicillin (2 g 4 hourly) 4-6 weeks, and
    - Gentamicin (1mg/kg 8-12 hourly)
  - Ampicillin resistant
    - Vancomycin(1g 12hourly) 4-6 weeks, and
    - Gentamicin (1mg/kg 8-12 hourly)

- **Suspect staph. endocarditis**
  - **Methicillin-susceptible staph:**
    - **Benzyl penicillin 1.2 g/4 hour or Flucloxacillin 2 g /4 hour IV +gentamycin**
  - **Methicillin-resistant strains of S aureus or penicillin resistant: (cardiac surgery):**
    - **Vancomycin 1g /12 hourly plus gentamycin 80-120mg 8hourly**

Treatment of endocarditis caused by HACEK:

- **Ceftriaxone 2 gm once IV or IM**
- **Ampicillin/sulbactam 1.2 gm/ 24h IV**
- **Gentamicin and vancomycin need to be Monitored to ensure adequate therapy and prevent toxicity.**

Surgery

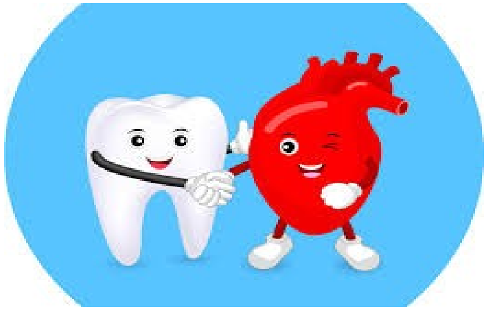
Indications

- ✓ ~~patients with direct extension of infection to~~ myocardial structuires.
- ✓ Prosthetic valve dysfunction.
- ✓ Congestive heart failure.
- ✓ Badly damaged valves.
- ✓ IE caused by fungi or gram-ve or resistant organisms.
- ✓ Large vegetations on echocardiography
- ✓ Recurrent embolic attacks.

Dental treatment and risk for IE

- **Oral commensal bacteria is an etiologic agents of IE.**
  - **Incidence of bacteremia for tooth extraction ranges from 18-85%, for local anesthesia is 73%, for periodontal surgery from 60- 90% and for tooth brushing or irrigation from 7-50%.**
- **Antibiotic prophylaxis (AP) is no longer mandatory for IE due to its side effect or complications as**
  - **Anaphylaxis, resistant bacteria.**
  - **Deaths 5-6 time from anaphylaxis more than IE.**

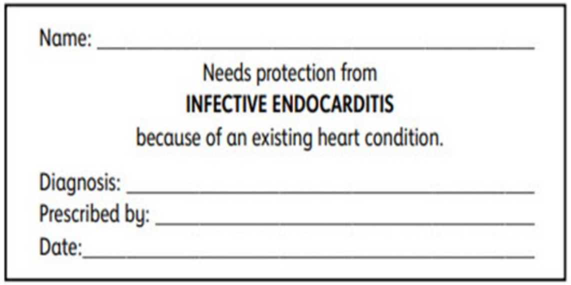
- **Decrease need of dental intervention with eliminate the necessity of AP in a patients at risk of IE through:**
  - **Maintaining optimal and good oral hygiene by:**
    - **Cleaning your teeth regularly with fluoride toothpaste**
    - **Regular dental visits once or twice/year**
    - **Avoid sugary snacks and drinks**
    - **Intensive preventive dental care**
    - **Infection control**
    - **Pre-interventional mouth rinsing with 0.2% chlorhexidine, iodine compounds and diluted oxygenated water**



- **UK National Institute for Health and Clinical Excellence (NICE) recommendations (2008):**
  - **No need for AP prior to dental manipulation in patients with a history of rheumatic fever or rheumatic valvular disease but still benefit in patients with prosthetic valve.**
- **NICE Clinical Guidelines were updated in 2015:**
  - **NICE did not change their recommendations (no AP in patients at high-risk of IE who are undergoing high-risk dental procedures).**

- **European Society of cardiology (ESC) (2023) and American Heart Association (AHA) recommends AP single dose 30-60 minutes before dental procedures for patients with cardiac conditions that involve:**
  - **Manipulation of gingival (gum) tissue.**
  - **Periapical region (area around the roots) of teeth.**
  - **Perforation of the oral mucosa.**

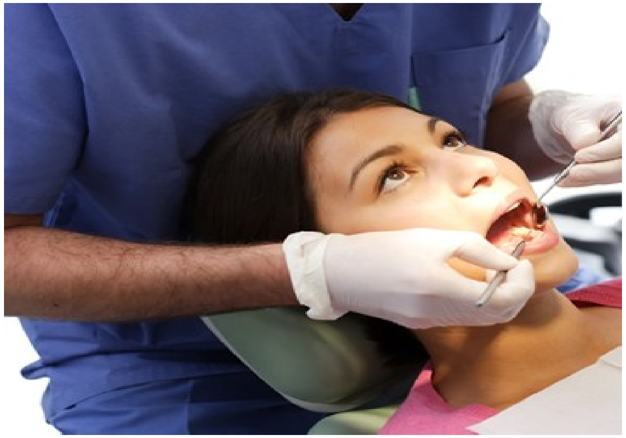
- **AP not suggested**
  - **Anesthetic injections through non-infected tissue.**
  - **Dental radiograph.**
  - **Placement of removable prosthodontic or orthodontic appliances.**
  - **Adjustment of orthodontic appliances**
  - **Placement of orthodontic brackets**
  - **Losing or shedding deciduous teeth**
  - **Bleeding from trauma to the lips or oral mucosa**



Name: ______________________

Needs protection from
INFECTIVE ENDOCARDITIS
because of an existing heart condition.

Diagnosis: ______________________
Prescribed by: ______________________
Date: ______________________

Invasive dental procedures	Non-invasive dental procedures
• Placement of matrix bands • Placement of sub-gingival rubber dam clamps • Sub-gingival restorations including fixed prosthodontics • Endodontic treatment before apical stop has been established • Preformed metal crowns (PMC/SSCs) • Full periodontal examinations (including pocket charting in diseased tissues) • Root surface instrumentation/sub-gingival scaling • Incision and drainage of abscess • Dental extractions • Surgery involving elevation of a muco-periosteal flap or muco-gingival area • Placement of dental implants including temporary anchorage devices, mini-implants • Uncovering implant sub-structures	• Infiltration or block local anaesthetic injections in non-infected soft tissues • BPE screening • Supra-gingival scale and polish • Supra-gingival restorations • Supra-gingival orthodontic bands and separators • Removal of sutures • Radiographs • Placement or adjustment of orthodontic or removable prosthodontic appliances 
N.B. In addition, antibiotic prophylaxis is not recommended following exfoliation of primary teeth or trauma to the lips or oral mucosa.	

- **Antibiotic regimens for IE prophylaxis are against S viridans.**
  - **AP is not recommended for bronchoscopy while recommended in invasive respiratory tract procedures (drainage of abscess).**
  - **Single dose 30 min. IV or 1 hour oral before procedure:**
    - **Oral amoxicillin 2 g PO, ampicillin 2g IV/IM.**
    - **Allergic to penicillin: erythromycin (Clindamycin: 600 mg PO or IV).**
      - **Clindamycin may cause frequent and severe reactions than other antibiotics.**
      - **AHA recommends azithromycin, clarithromycin, or doxycycline be used as alternatives in individuals who are allergic to penicillin/ ampicillin.**

Situation	Agent and dose
Oral	Amoxicillin 2 gm
Unable to take oral medication	Ampicillin 2 g IM or IV or Cefazolin or Ceftriaxone 1 g IM or IV
Allergic to penicillin or ampicillin—oral	Cephalexin 2 g or Azithromycin or Clarithromycin 500 mg Doxycycline 100 mg
Allergic to penicillin/ ampicillin and unable to take oral medication	Cefazolin or Ceftriaxone 1 g IM or IV

- **Notes:**
  - **If a patient received a course of antibiotics in the 6 weeks prior to the procedure, chose a different antibiotic class for prophylaxis**
  - **Clindamycin is no longer recommended for prophylaxis before dental procedure.**
  - **Cephalosporin should not be used in an individual with a history of anaphylaxis, angioedema, or urticarial with penicillin or ampicillin.**

- Patients with high cardiac risk who undergo a surgical procedure (infected skin, skin structure, or musculoskeletal tissue) should receive antibiotic against staphylococci and beta-hemolytic streptococci:
  - Penicillin, cephalosporin.
- If unable to tolerate beta-lactam antibiotics, or methicillin-resistant strains of S. aureus:
  - Vancomycin.

- **Antibiotics are no longer recommended for endocarditis prophylaxis for patients undergoing genitourinary or gastrointestinal tract procedures.**

- **Which of the following organisms is the least implicated in infective endocarditis?**
- **A. Candida species.**
- **B. Streptococcus species.**
- **C. Staphylococcus species.**
- **D. Enterococcus species.**

- **A 65-year-old male developed endocarditis with viridans streptococci after dental treatment. He has no history of drug allergy. Which of the following intravenous regimens is most appropriate?**
- **A. Erythromycin 1hour before procedure.**
- **B. Penicillin G for 4 weeks.**
- **C. Ampicillin/sulbactam.**
- **D. Vancomycin + gentamicin for 2 weeks.**

- **Which of the following diagnostic criteria for infective endocarditis?**
- **A. Positive blood cultures and echocardiographic changes.**
- **B. Echocardiography and plain x- ray chest.**
- **C. ECG changes and positive physical findings.**
- **D. Positive physical findings and positive blood cultures.**

- **Which of the following is a complication of infective endocarditis?**
- **A. Embolization.**
- **B. Pneumonia.**
- **C. Diarrhea.**
- **D. Sinusitis.**


THANK YOU